Public Summary of Training Content for General-Purpose AI models

This template is provided by the European Commission and required to be filled in by providers of general-purpose AI models prior to their placing on the Union market in order to comply with their obligation under Article 53 (1)(d) of Regulation (EU) 2024/1689 (AI Act).
For more information and guidance see Commission's Explanatory Notice and Template for the Public Summary of Training Content for general-purpose AI models | Shaping Europe's digital future.

Version of the Summary:	*v1.5*
Last update:	*6.10.2026*

1. General information

1.1. Provider identification

Provider name and contact details:	***Swiss National AI Institute*** *% EPFL AI Center* *ELE 136, Station 11* *CH-1015 Lausanne, Switzerland* *Website: https://apertus-ai.org* *General contact: llm-requests@swiss-ai.org*
Authorised representative name and contact details:	*Not applicable pursuant to Article 54(6) of Regulation (EU) 2024/1689. The models covered by this Summary are released under a free and open-source licence permitting access, use, modification and distribution. Their weights, architecture information and usage information are publicly available, and they are not general-purpose AI models with systemic risk.*

1.2. Model identification

Versioned model name(s):	*Apertus-v1.5-8B* *Apertus-v1.5-70B* *https://huggingface.co/collections/swiss-ai/apertus-v15* *This Summary covers two model versions, whose training content is similar. They are reported together in accordance with point (30) of the Commission Explanatory Notice to the Template.*
Model dependencies:	*Apertus-8B-2509* *Apertus-70B-2509* *https://huggingface.co/collections/swiss-ai/apertus-v1* *Apertus 1.5 is a continued pre-training on top of the previous generation of the same model. These are all decoder-only transformer text-generation models released under the Apache 2.0 license. The model card for each is published at the URLs above.*
Date of placement of the model on the Union market:	*Both models were publicly announced on July 24, 2026. Neither has been further trained since its release, and no new version has been placed on the market. Any revision to tokenizer files will not alter the model weights or the training data.*

1.3 Modalities, overall training data size and other characteristics

This Section requires general information about the overall training data after pre-processing and before the training of the model.

Modality	Training data size	Types of content
☒ Text	☐ Less than 1 billion tokens ☐ 1billion to 10 trillions tokens ☒ More than 10 trillions tokens Alternatively, specify the approximate size in a different measurement unit: *17 trillion tokens (70B)* *19 trillion tokens (8B)*	*Public text-only datasets derived mainly from web documents written in over 1000 languages, while respecting the consent of right holders. We ensure training data is fully transparent and reproducible, so as to make Apertus an open-data and open-weights model. Data is filtered for compliance and personal identification information when applicable.*
☒ Image	☐ Less than 1 million images ☐ 1Million to1 billion images ☒ More than 1 billion images	*Public image-only and image-text-pairs, gathered from permissive datasets, themselves derived mainly from open image repositories and web data. Data is filtered for compliance and personal identification information when applicable.*
☒ Audio	☐ Less than 10 000 hours ☒ 10 000 to 1 million hours ☐ More than 1 million hours	*Public audio-only and audio-transcripts-pairs, gathered from permissive datasets, themselves derived mainly from open initiatives (such as commonvoice) and web data.*
☐ ~~Video~~	☐ Less than 10 000 hours ☐ 10 000 to1 million hours ☐ More than 1 million hours	
☐ ~~Other~~	*Specify the modality and for each one indicate approximate size and unit of measurement*	

Latest date of data acquisition/collection for model training:	*Pre-training dataset knowledge cutoff is April 28th, 2026.*
Description of the linguistic characteristics of the overall training data:	***Languages**. The pretraining dataset includes more than **1000 languages** (1782 language-script pairs), as provided by the FineWeb-2 and FineWeb-2-HQ datasets respectively. The amount of data per language reflects the natural frequency of web data in each language, thus improving representation of many communities with languages not present in most leading LLMs yet.* ***Domain-specificity.** Emphasis has been placed on **legal** (with datasets such as Multi-Legal-Pile or Swiss-caselaw), **medical** (with datasets such as MedTrinity), as well as **science-related** (finemath) contents for this release.*
Other relevant characteristics of the overall training data:	*Selected datasets (or dataset subsets) adhere to our strict policy of full license permissiveness (excluding sharealike, non-commercial and any other restrictive licenses). Data has been rigorously filtered for respecting consent by website owners (opt-out for AI crawlers, also retroactively), remove PII, remove toxic content, and avoid verbatim memorization during model training (see below).*
Additional comments (optional):	*The Apertus tokenizer builds upon the Mistral v3 (tekken), and is used for data size statistics (see also the Apertus technical report).*

2. List of data sources

This Section requires information about specific sources of data used to train the general-purpose AI model. In this section "dataset" should be understood as a single, pre-packaged collection of data. The filtering and pre-processing of data collected from the same pre-packaged collection should not be considered a new dataset to be disclosed separately in the sections below. If a particular dataset can be assigned to more than one of the categories below, providers should select the most relevant category and only report the dataset in that category, except in the case of synthetic data (see Section 2.5).

2.1. Publicly available datasets

This Section requires information about datasets that were used to train the model and which have been compiled by a third party, are made available publicly for free, and are readily downloadable as a whole or in predefined chunks, such as datasets and collections available on public repositories and online platforms, specialised websites, or snapshots of common crawl. The public availability of the datasets for free does not mean that the content at issue is necessarily free of rights since it may be subject to licensing arrangements or conditions of use (e.g., certain free and/or open licenses may determine the scope of the uses, including prohibiting uses relating to model training).

A dataset is considered to be "large" if the total data size for any one of the modalities contained in the dataset exceeds 3% of the size of all publicly available datasets for that modality used for training. The size of the dataset should be based on its size after pre-processing (for example filtering), and without splitting the dataset to prevent reporting circumvention.

Have you used publicly available datasets to train the model?

☒ Yes ☐ No

If yes, specify the modality(ies) of the content covered by the datasets concerned:

☒ Text ☒ Image ☐ Video ☒ Audio

☐ Other *If so, please specify…*

List of large publicly available datasets:

The following large pretraining datasets were not used as is, but were additionally filtered for opt-out retrospectively, for toxicity, high quality, and other preprocessing as detailed below.

The same versions and filtering techniques were used for the datasets used both in Apertus v1 and v1.5[1] and are detailed in our first technical report. For datasets used only in v1.5, filtering and pre-processing was very similar to v1. We used the same codebase[2] and the same list of robots.txt. The same PII-removal filter was used. Slight overhauls mainly concerned technical implementation details. For robots-filtered image datasets (MINT-1T, pixmo-cap), we only excluded the sources that were within our list and had robots.txt indications (i.e. not excluding sources that were not gathered by our list).

With respect to training from a purely technical perspective, the most important change is naturally the addition of multimodal data.

In addition to scale, we focused on data quality through curation, deduplication, educational filtering and domain relevance. Our resulting data-mixture supports robust, general-purpose, and specialised capabilities while advancing open and responsible AI development.

[1] The following datasets are concerned: dclm-edu, fineweb-2, finetranslations, finemath, MegaMath, stack-edu.
[2] Public version accessible at: github.com/swiss-ai/pretrain-data

Text Datasets

HuggingFaceFW/fineweb-2: Large-scale, high-quality multilingual web text corpus derived from filtered Common Crawl data. Serves as a primary general-purpose pretraining source for LLMs, with strong emphasis on diversity and reduced noise. License: ODC-BY. (Subject to PII removal and robots.txt filtering.)

HuggingFaceTB/dclm-edu: Curated educational web dataset from the DataComp-LM project. Provides high-signal, knowledge-rich text optimised for reasoning and factual learning in language models. License: CC-BY-4.0. (PII removal and robots.txt filtering applied.)

nvidia/Nemotron-CC-v2.1: NVIDIA-curated Common Crawl dataset (v2.1) featuring quality scoring and filtering for large-scale LLM pre-training. License: Nvidia's custom data and model license (permissive, see dataset page for terms).

HuggingFaceFW/finepdfs-edu: High-quality collection of educational, scientific, and technical PDF documents with extracted clean text. Enhances long-context and domain knowledge capabilities. License: ODC-BY. (Filtered for quality and compliance.)

joelniklaus/Multi_Legal_Pile: Specialized multilingual legal corpus aggregating statutes, case law, contracts, and regulatory texts from multiple jurisdictions. Strengthens legal reasoning and domain adaptation. License: only compliant (non-SA, non-NC) subsets of this compound dataset were used. (PII removal and robots.txt filtering applied.)

nvidia/Nemotron-Pretraining-Code-v1: Large-scale, curated code corpus from NVIDIA designed to boost programming, software engineering, and logical reasoning abilities.License: Nvidia's custom data and model license (permissive, see dataset page for terms).

Apertus v1.5 Preference data: the alignment mixture used for the offline DPO stage of Apertus v1.5 training, applied to the 70B model, with prompts from Ai2's Olmo 3 Dolci-Instruct-DPO dataset.

Apertus v1.5 SFT mixture: The dataset mixture used for the final Supervised Finetuning Stage run of Apertus v1.5 models (data sources and distribution are stated in the Technical Report and dataset card)

Audio Datasets

mozilla/CommonVoice24: Mozilla's crowdsourced multilingual speech corpus with validated transcriptions across many languages and accents. A cornerstone dataset for inclusive, robust automatic speech recognition (ASR) and text-to-speech (TTS). License: CC-BY-1.0.

speechcolab/gigaspeech: Large-scale English speech recognition corpus (~10k hours) sourced from audiobooks, podcasts, and YouTube with high-quality transcriptions. Supports general-domain ASR and audio understanding. License: Apache-2.0.

MLCommons/peoples_speech: One of the largest publicly available multilingual speech datasets, containing tens of thousands of hours of diverse, real-world speech. License: CC-BY-SA / CC-BY.

facebookresearch/voxpopuli: Large multilingual speech corpus extracted from European Parliament sessions, covering numerous EU languages with aligned transcripts. Excellent for cross-lingual and parliamentary-domain speech tasks. License: CC-BY-1.0.

k2-fsa/libriheavy: Massive clean English read-speech corpus (tens of thousands of hours) built on LibriSpeech audiobooks with precise alignments. Known for high acoustic quality and utility in ASR/TTS research. License: Apache-2.0.

facebook/omnilingual-asr-corpus: Massive multilingual speech corpus spanning a wide range of languages and dialects, created to advance open ASR systems globally. License: CC BY 4.0.

nvidia/Granary: a large-scale, open-source multilingual speech dataset covering 25 European languages for Automatic Speech Recognition (ASR) and Automatic Speech Translation (AST) tasks.

Image Datasets

mlfoundations/MINT-1T: Landmark large-scale image-text dataset scaled to trillions of tokens, specifically designed to dramatically expand open-source multimodal pretraining data. License: CC-BY-4.0. (Includes PII removal and robots.txt filtering.)

UCSC-VLAA/Recap-DataComp-1B: Billion-scale image-text dataset featuring high-quality recaptions of the original DataComp-1B collection. Optimized for vision-language pretraining and detailed visual understanding. License: CC-BY-4.0.

dclure/laion-aesthetics-12m-umap: Curated 12-million-image subset of LAION focused on high aesthetic quality (via CLIP and aesthetic scoring). Popular for training visually pleasing image understanding and generation models. License: MIT.

mvp-lab/LLaVA-OneVision-1.5-Mid-Training-85M: Large-scale (85M samples) mid-training dataset for vision-language models, emphasizing instruction tuning and multimodal alignment. License: Apache-2.0.

UCSC-VLAA/MedTrinity-25M: Large medical image-text dataset (25M samples) with rich annotations and captions. Key resource for building specialized medical vision-language understanding and diagnostic assistance features. License: only compliant (non-SA, non-NC) subsets of this compound dataset were used (see full list on the dataset's page).

common-canvas/commoncatalog-cc-by:a large collection of high-resolution (up to 4k) Creative Common images collected in 2014 from Yahoo and Flickr, captioned by users.

Spawning/pd12m-full: a large public domain image-text dataset, with sufficient size to train foundation models while minimizing copyright concerns and community-driven dataset governance mechanisms that reduce harm and support reproducibility over time.

General description of other publicly available datasets not listed above:

Other smaller publicly available and permissively licensed datasets were used for the purposes described above. The exhaustive list of pre-training datasets is accessible as a versioned csv on our dedicated GitHub Repository. Supervised fine-tuning and alignment data are publicly accessible here and here respectively. Generally, all republished datasets used for Apertus v1.5 will be made publicly available in the swiss-ai collection on Hugging Face.

Additional comments (optional):

Training data preparation code (for pre- and post-training) is entirely reproducible and is or will be made publicly available in the following repositories:

- *Version 1.0 text pretraining data preparation pipelines*
- *Version 1.5 text posttraining data preparation pipelines*
- *Version 1.5 multimodal data preparation pipelines*

Most vision datasets include image bytes directly, but some provide only URLs requiring redownload. Successful retrieval rates varied.
For wild URLs:

- *dclure/laion-aesthetics-12m-umap (52.82%)*
- *mlfoundations/MINT-1T-HTML (64.10%)*
- *UCSC-VLAA/Recap-DataComp-1B (72.49%)*
- *Salesforce/blip3-grounding-50m (64.43%)*
- *OpenFace-CQUPT/FaceCaption-15M (72.50%)*
- *YangQiee/HQ-50K (72.02%)*
- *allenai/pixmo-point-explanations (77.36%)*
- *allenai/pixmo-ask-model-anything (81.74%)*
- *allenai/Molmo2-MultiImageQA (81.63%)*
- *allenai/pixmo-cap-qa (89.23%)*

For CDN/concentrated-host sources:

- *bitmind/open-images-v7 (100.00%)*
- *madebyollin/megalith-10m (97.82%)*
- *allenai/pixmo-cap (98.53%)*

2.2 Private non-publicly available datasets obtained from third parties

This Section requires information about private non-publicly available datasets of third parties that are not publicly available and not disclosed under Section 2.1. These include:

1) *datasets for which transactional commercial licensing agreements were concluded between the provider and the rightsholders or their representatives, including by collective management organisations and legitimate content aggregators who have the right to collectively license works on behalf of rightsholders (Section 2.2.1);*

2) other private datasets obtained through data intermediaries, non-publicly available databases and datasets of third parties for which transactional commercial licenses have not been concluded with rightsholders or their representatives (Section 2.2.2).

2.2.1. Datasets commercially licensed by rightsholders or their representatives

Have you concluded transactional commercial licensing agreement(s) with rightsholder(s) or with their representatives?

☐ Yes ☒ No

If yes, specify the modality(ies) of the content covered by the datasets concerned:

☐ **Text** ☐ **Image** ☐ **Video**

☐ **Audio** ☐ **Other** ***If so, please specify…***

2.2.2. Private datasets obtained from other third parties

Have you obtained private datasets from third parties that are not licensed as described in Section 2.2.1, such as data obtained from providers of private databases, or data intermediaries?	☐ Yes ☒ No
If yes, specify the modality(ies) of the content covered by the datasets concerned:	☐ Text ☐ Image ☐ Video ☐ Audio ☐ Other
If publicly known, list private datasets obtained from other third parties:	*N/A*
General description of non-publicly known private datasets obtained from third parties	*N/A*
Additional comments (optional):	*N/A*

2.3 Data crawled and scraped from online sources

This Section requires information about crawled, scraped data, or otherwise compiled from online sources directly by the provider of the model or on their behalf (i.e. excluding publicly available datasets already compiled by third parties and made available on platforms such as common crawl that are covered under Section 2.1).

Were crawlers used by the provider or on behalf of?	☒ Yes ☐ No
If yes, specify crawler name(s)/identifier(s):	*"Apertus/1.0" is used as an identifier for research data preparation where possible.*
Purposes of the crawler(s):	*Retrieve data in a responsible, opt-out compliant way.*
General description of crawler behaviour:	*Custom crawlers are used on specific datasets to fetch multimodal content, such as images or PDF documents. Crawlers include respect of Captchas, password protected websites, paywalls, and robots.txt rules.*
Period of data collection:	*From 01/2026 to 04/2026*

Comprehensive description of the type of content and online sources crawled:	*While the vast majority of the Apertus training data is from pre-existing public datasets (such as from Common Crawl), additional Image-text content was scraped from public institutional sources: NASA science and aeronautics imagery, Smithsonian cultural-heritage collections, Swiss federal cartographic and geospatial data, and Our World in Data articles.* *Data sources were US/Swiss government data and educational/research services, accessed through public APIs, bulk release or web services. Swiss maps contain German, French and Italian labels; other content is predominantly English. The collections are topic-based, although cultural-heritage material may depict people and historical communities. The resulting datasets are:* *https://huggingface.co/datasets/swiss-ai/nasa* *A public-domain subset of the NASA Image and Video Library, filtered and captioned with a Qwen vision-language model.* *https://huggingface.co/datasets/swiss-ai/owid* *Chart images, short data-insight posts, and narrative articles from Our World in Data, prepared as image-text data.* *https://huggingface.co/datasets/swiss-ai/smithsonian* *Museum objects from Smithsonian with captions grounded in the museum's catalogue record.* *https://huggingface.co/datasets/swiss-ai/swisstopo* *Map tiles rendered from the Swiss Federal Geoportal's public WMS service with captions.*
Type of modality covered:	☒ Text ☒ Image ☐ Video ☒ Audio ☐ Other *If so, please specify…*
Summary of the most relevant domain names crawled:	*Content was extracted from identified public institutional collections and services, rather than through general-purpose web crawling. Primary content-source domains were:* - *si.edu,* - *nasa.gov,* - *geo.admin.ch,* - *ourworldindata.org.* *Smithsonian bulk content was obtained through its official AWS Open Data delivery channel (amazonaws.com).*
Additional comments (optional):	*Source code of multimodal crawlers can be found at https://github.com/swiss-ai/multimodal-data*

2.4 User data

This Section requires information about user data collected by all services and products of the provider, including through mail services, social media platforms, content platforms or interaction with the providers' AI models and/or systems. This does not cover data licensed by users based on commercial transactional agreements described in Section 2.2.1., or customer data to fine-tune models for specific purposes.

Was data from user interactions with the AI model (e.g. user input and prompts) used to train the model?	☐ Yes ☒ No
Was data collected from user interactions with the provider's other services or products used to train the model?	☐ Yes ☒ No
If yes, provide a general description of the provider's services or products that were used to collect the user data:	*N/A*
Type of modality covered:	*N/A*
Additional comments (optional):	*N/A*

2.5 Synthetic data

This Section requires information about synthetic data created by or on behalf of the provider for training the model directly on the outputs of another AI model, in particular through model distillation or model alignment (e.g. AI feedback through reinforcement learning). This does not include the use of AI models to clean or enrich data (e.g. AI-generated metadata to enrich or modify a dataset, such as creating depth maps or text descriptions of images). In case this concerns publicly available datasets as described in Section 2.1, these should be reported in that Section of the Template. In case this concerns synthetic datasets created by third parties on behalf of the provider, these should be reported in this Section of the Template instead of in Section 2.2.2.

Was synthetic AI-generated data created by the provider or on their behalf to train the model?	☒ Yes ☐ No
If yes, modality of the synthetic data:	☒ Text ☐ Image ☐ Video ☐ Audio ☐ Other *If so, please specify…*
If yes, specify the general-purpose AI model(s) used to generate the synthetic data if available on the market:	*Qwen3.6-27B and Qwen3.5-397B-A17B were used across multiple vision datasets to generate synthetic instruction, question-answer and grounded response text paired with existing, non-synthetic images.* *The following models were used to generate synthetic post-training data: Qwen3-235B, Kimi-K2.5, MiniMax-M2.7, Qwen3.5-397B, GPT-OSS-120B-Safeguard.*

Information about other AI models, including provider's own AI model(s) not available on the market, used to generate synthetic data to train the model to which this Summary applies:	*N/A. No other non-market AI model was used to generate synthetic data.*
Additional comments (optional):	*In addition to the models listed above, publicly available models were also used for data processing and enrichment. These included* - *Qwen3.5-9B,* - *Qwen2.5-VL-72B-Instruct,* - *Gemma4-31B-IT,* - *Gemma4-26B-A4B-IT,* - *MedGemma1.5-4B-IT,* - *Kimi-VL-A3B-Instruct,* - *Kimi-VL-A3B-Thinking-2506,* - *and EuroLLM-22B-Instruct-2512.* *Their uses included recaptioning, caption cleaning, OCR, translation and quality assessment of existing source material.* *All models used in data generation were hosted on our own research infrastructure.*

2.6 Other sources of data

This Section requires information about data that does not fall under any of the categories in the previous Sections, for example data collected from offline sources, self-digitised media (e.g., digitised analog text context, images), datasets labelled by humans commissioned by the provider, or human generated data through reinforcement learning.

Have data sources other than those described in Sections 2.1 to 2.5 been used to train the model?	☐ Yes ☒ No
If yes, provide a narrative description of these data sources and the data:	N/A
Additional comments (optional):	*N/A*

3. Data processing aspects

3.1. Respect of reservation of rights from text and data mining exception or limitation

This Section concerns measures implemented by the provider to identify and comply with the reservation of rights from the text and data mining (TDM) exception or limitation expressed pursuant to Article 4(3) of Directive (EU) 2019/790, as outlined in the copyright policy put in place by the provider in accordance with Article 53(1)(c) AI Act.

Are you a Signatory to the Code of Practice for general-purpose AI models that includes commitments to respect reservations of rights from the TDM exception or limitation?	☐ Yes ☒ No
Describe the measures implemented before model training to respect reservations of rights from the TDM exception or limitation before and during data collection, including the opt-out protocols and solutions honoured by the provider or, as applicable, by third parties from which datasets have been obtained:	*We respect standard machine-readable opt-out by all websites. In addition, we remove data from websites which have recently opted out by specifying at least one of the common AI crawlers, at the time of January 2025. Crucially, we have applied such removals also retroactively in all earlier crawls since 2013, of each corresponding website present in our datasets. Pretraining and posttraining datasets were additionally filtered for licence compliance, and text data was processed with PII removal.*
Additional comments (optional):	*Our Acceptable Use Policy of Apertus LLM is a documented Usage Policy statement. A summary of our copyright policy can be found on the model card (Legal aspects), on our website, and in our Code of Practice.*

3.2 Removal of illegal content

This Section concerns measures taken to avoid or remove illegal content under Union law from the training data (such as blacklists, keywords, and model-based classifiers), without requiring disclosure of specific details about the provider's internal business practices or trade secrets. Such measures are advisable if the training data is likely to include illegal or unlawful content under Union law, in particular child sexual abuse material and terrorist content and the non-authorised use of material protected by intellectual property rights. Such measures do not include data selection practices, for example to increase the capability of the model.

General description of measures taken:

Before training, we remove toxic documents from the pretraining corpora, following the same approach as Apertus v1. We do so by employing a deep learning classifier trained on top of XLM-Roberta multilingual embeddings. The classifiers were trained using the multilingual datasets provided by Pleias/toxic-commons

In addition to toxicity filtering, most datasets also were filtered by additional quality classifiers, which we make transparently available, and which further help to reduce problematic content. Image dataset are deduplicated and converted to appropriate formats and dimensions, among other common preprocessing practices.

3.3. Other information (optional)

Other relevant information about data processing (optional):

In addition to the full transparency of ensuring all training data of our models is openly available and reproducible (Apertus models being open-data and open-weights), we also employed techniques to minimize memorization or potentially remaining copyrighted content: During training, we employ the Goldfish loss technique , which disables verbatim memorization of text sequences longer than 50 tokens.

More precisely, every 50th token (on average) of our pretraining data is not provided a prediction target, i.e. has no loss function, and thus breaks verbatim memorization beyond that sequence length. We provide more detailed results on the success of this mitigation technique in the model's technical report.